

SUPPLEMENTARY

Supplementary Material — Centrality-Based KV-Cache Eviction

Companion to the main report (REPORT.pdf): detailed per-press systems figures and per-task result tables. See the report §4.4–4.5 for the summary and interpretation.

S.1 Per-press systems figures

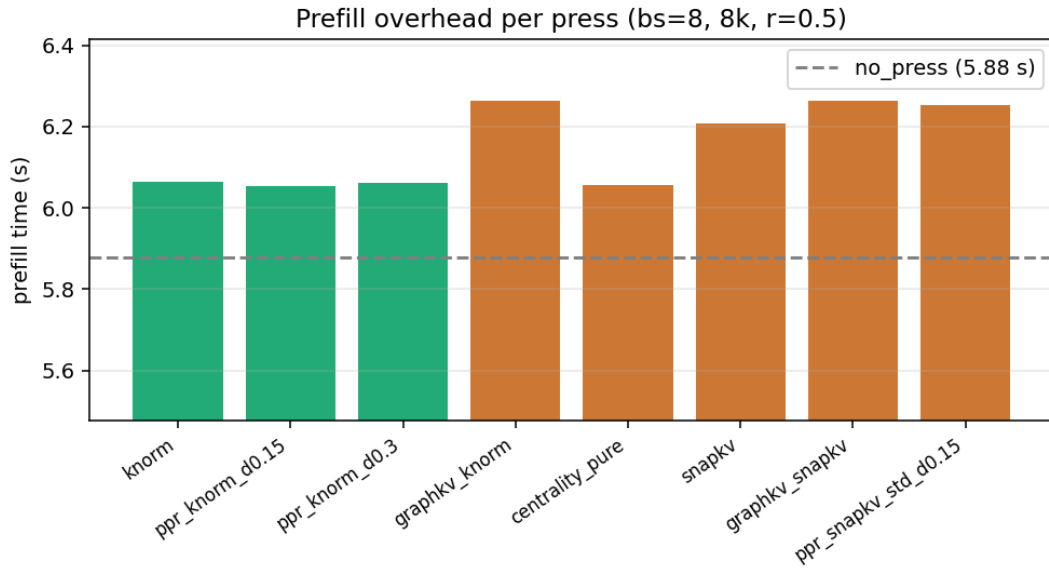


Figure S1: Prefill scoring overhead — the only press-dependent systems cost; $ppr_knorm \approx knorm$, cheaper than $graphkv$ / $SnapKV$ -based presses.

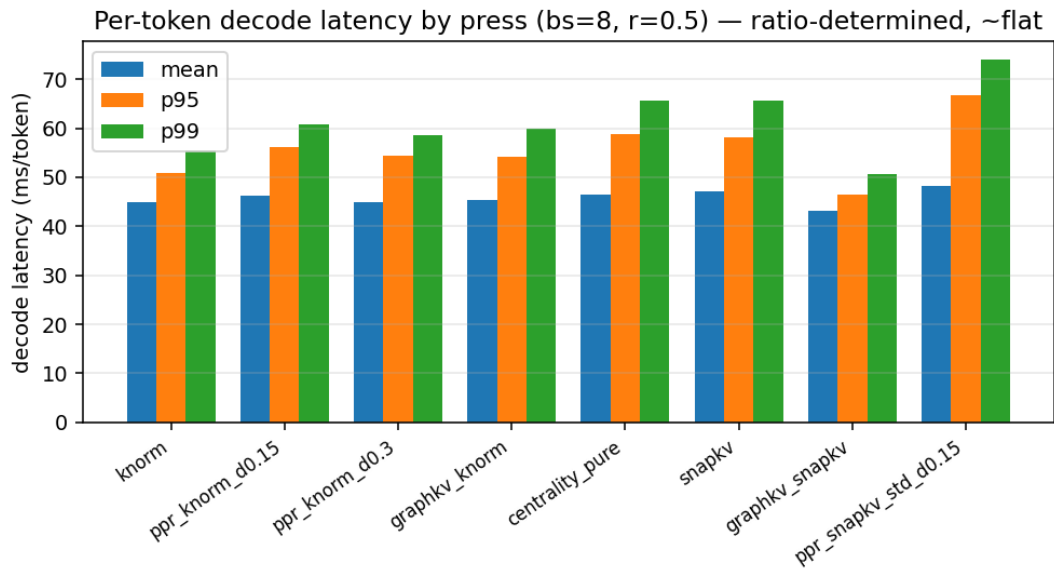


Figure S2: Per-token decode latency (mean/p95/p99) by press at $r=0.5$ — ratio-determined, ~flat.

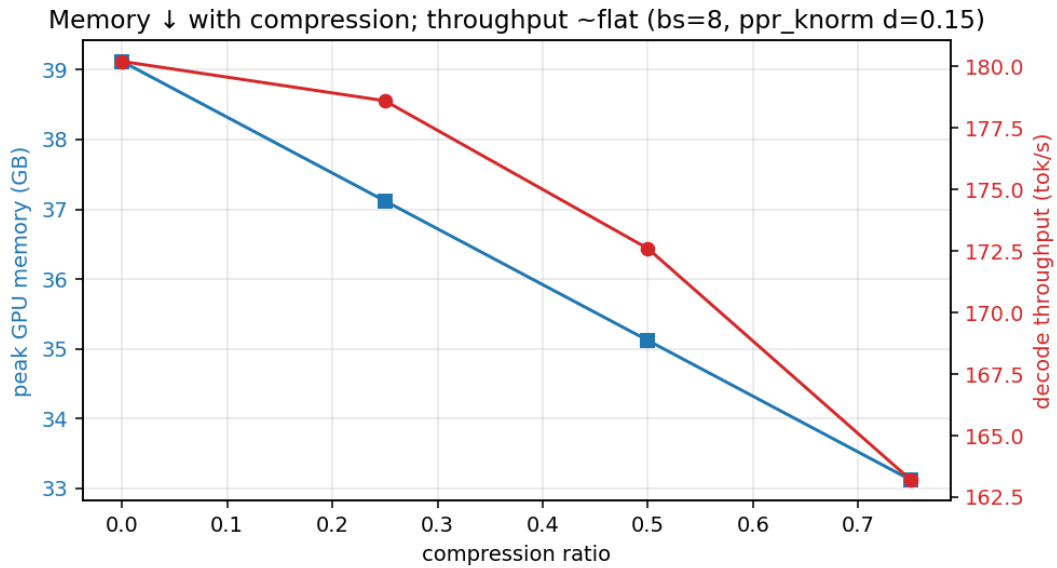


Figure S3: Peak GPU memory falls linearly with compression; decode throughput ~flat at this scale.

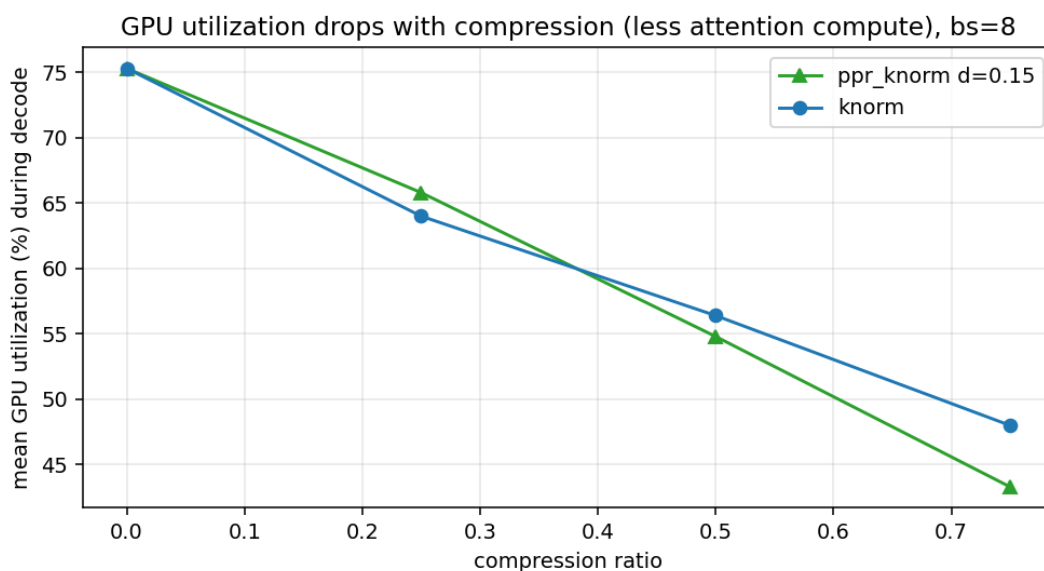


Figure S4: Mean decode GPU utilization drops with compression (less attention compute per token).

S.2 Per-task tables

RULER 4096 @ compression 0.5 (string-match %, usable tasks)

task	no_press	knorm	graphkv	snapkv	ppr d=0.15	pure
niah_single_1	100	100	100	94.7	100	0
niah_single_2	100	80.0	20.0	97.1	100	0
niah_single_3	100	2.9	5.9	0	73.5	0
niah_multikey_1	100	30.0	40.0	90.0	90.0	6.7
niah_multikey_2	100	4.8	81.0	61.9	100	0
niah_multikey_3	100	4.5	36.4	22.7	18.2	0
qa_2	58.1	32.3	48.4	48.4	48.4	29.0

LongBench multi-hop @ compression 0.5 (F1 %)

subtask	no_press	knorm	graphkv	snapkv	ppr d=0.15
hotpotqa	58.5	54.1	45.9	56.7	49.2
2wikimqa	53.1	45.9	46.7	53.5	53.4
musique	30.7	19.5	30.9	25.2	23.4

Per-run configs + metrics are under project/reproduction/results/
 (config.yaml + metrics.json each; the aggregated per-example *_long.csv are
 report-only — regenerable, see project/reproduction/MANIFEST.md).
 Regenerate figures with analysis/make_figures.py and paired stats with
 analysis/analyze_paired.py.

S.3 Iso-accuracy operating-point curve (16k, batch 8)

16k / batch 8: memory scales linearly with cache; speed moves only past the knee

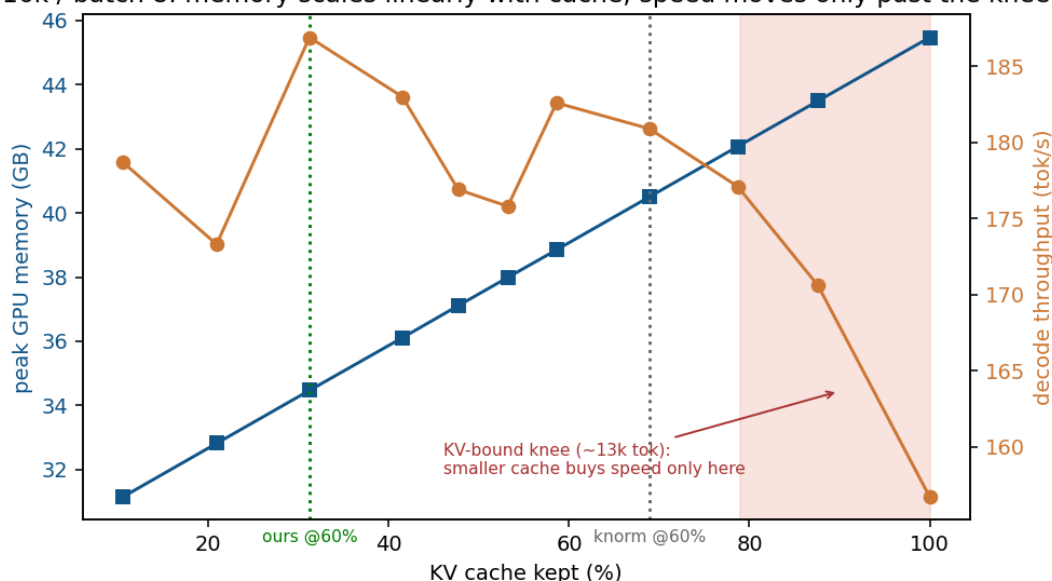


Figure S5: Directly measured peak GPU memory and decode throughput vs. kept-cache fraction (16k prompt, batch 8, analysis/iso_systems.py on results/results_iso_systems_16k_bs8.csv). Peak memory falls linearly with the cache; throughput is essentially flat once the cache is compressed (decode is weight-bound at 8B/16k/bs8). The dotted lines mark the iso-accuracy operating points for a 60 % RULER target: the base press must keep ~69 % of the cache, ours only ~31 % — same accuracy, $2.2\times$ less cache and ~6 GB less peak memory, while throughput is unchanged (the speed win needs a more KV-bound regime).

Appendix — Code & data artifacts

- **Press code:** kvpress/presses/centrality_press.py; registration in kvpress/__init__.py, evaluation/evaluate_registry.py, tests/default_presses.py, README.md; unit test in tests/presses/test_centrality_press.py.
- **GraphKV comparison baseline (project/additional_benchmarks/):** decay_propagation_press.py, test_decay_propagation_press.py, run.py (runtime-injected registry entry), README.md.
- **Benchmark + analysis (project/reproduction/):** scripts/ruler_rerun.py, scripts/sweep_longbench.py, scripts/tau_sweep.sh, scripts/submission_grid_llama.sh, scripts/submission_grid_qwen.sh, scripts/fetch_eval_datasets.sh, scripts/eval_cfg_dec.yaml; analysis/analyze_paired.py, analysis/make_figures.py, analysis/speed_memory.py, analysis/iso_systems.py, analysis/recall_probe.py, analysis/rank_vs_board.py, analysis/teleport_entropy.py, analysis/kvpress_leaderboard_raw.csv.
- **Data/logs (project/reproduction/results/):** board_grid/, ruler_screen/, and ablations/ run directories (each with config.yaml +

metrics.json; large predictions.csv omitted); figures in report/figures/ (fig1–fig12).

- **Reproducibility:** requirements.txt, Dockerfile, README.md (how-to-benchmark), MANIFEST.md (report number → run dir), BASELINE.md, REGENERATE.md, .github/workflows/centrality-ci.yml (CI: comparison-benchmark tests on every push).